We create chemistry

Safety data sheet

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 11.05.2016 Version: 1.1
Product: **Oily Vitamin Blend 02-118**
(ID no. 30041068/SDS_GEN_00/EN)
Date of print 12.07.2025

1. Identification

Product identifier

Oily Vitamin Blend 02-118

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Vitamin

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Nutrition and Health

Telephone: +49 621 60-48434
E-mail address: EN-global-safety-data@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Skin Corr./Irrit. 3
Repr. 1B (unborn child)
Aquatic Chronic 4

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H316	Causes mild skin irritation.
H360	May damage the unborn child.
H413	May cause long lasting harmful effects to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves/protective clothing/eye protection/face protection.
P273	Avoid release to the environment.
P202	Do not handle until all safety precautions have been read and understood.

Precautionary Statements (Response):

P308 + P311	IF exposed or concerned: Call a POISON CENTER or doctor/physician.
P332 + P313	If skin irritation occurs: Get medical advice/attention.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container to hazardous or special waste collection point.

Other hazards

According to UN GHS criteria

When finely distributed, self-ignition is possible. The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

vitamin blend, Retinyl propionate, colecalciferol; Vitamin D3

dissolved in: Sunflower oil

Hazardous ingredients (GHS)
According to UN GHS criteria

Retinyl propionate

Content (W/W): >= 25 % - < 50 %	Acute Tox. 5 (oral)
CAS Number: 7069-42-3	Skin Corr./Irrit. 3
EC-Number: 230-363-2	Repr. 1B (unborn child)
	Aquatic Chronic 4
	H316, H303, H360, H413

colecalciferol; Vitamin D3

Content (W/W): >= 0,2 % - < 0,3 %	Acute Tox. 2 (Inhalation - dust)
CAS Number: 67-97-0	Acute Tox. 2 (oral)
	Acute Tox. 2 (dermal)
	STOT RE (Kidney, Blood vessel system) 1
	M-factor acute: 1
	M-factor chronic: 1
	H310, H330, H300, H372

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed
Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11.

Indication of any immediate medical attention and special treatment needed
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media
Suitable extinguishing media:
water spray, dry powder, carbon dioxide, foam

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture
harmful vapours
Evolution of fumes/fog.

Advice for fire-fighters
Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
Cool endangered containers with water-spray.

6. Accidental Release Measures

High risk of slipping due to leakage/spillage of product.

Personal precautions, protective equipment and emergency procedures
Wear suitable protective clothing. Information regarding personal protective measures see, section 8.
Avoid contact with the skin, eyes and clothing.

Environmental precautions
Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up
For small amounts: Pick up with suitable absorbent material.
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations.

Reference to other sections
Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

7. Handling and Storage

Precautions for safe handling
Handle in accordance with good industrial hygiene and safety practice. Avoid contact with the skin, eyes and clothing. Wear suitable protective clothing and gloves.

Protection against fire and explosion:
Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy. Risk of self-ignition when a large surface area is produced due to fine dispersion.

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE)
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place. Protect from air. Protect contents from the effects of light.

Specific end use(s)
For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

8001-21-6: Sunflower oil

Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:
Suitable chemical resistant safety gloves (EN 374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc. Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Avoid contact with skin. Do not inhale gases/vapours/aerosols. Wearing of closed work clothing is recommended. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid
Colour:	yellow
Odour:	slight odour
Odour threshold:	Not determined due to potential health hazard by inhalation.
pH value:	insoluble
Melting point:	The substance / product decomposes therefore not determined., The data given are those of the active ingredient.
Boiling point:	The substance / product decomposes therefore not determined., The data given are those of the active ingredient.
Flash point:	> 128 °C

Evaporation rate:
negligible
Flammability: hardly combustible
Lower explosion limit:
For liquids not relevant for classification and labelling.
Upper explosion limit:
For liquids not relevant for classification and labelling.
Ignition temperature: > 300 °C
Vapour pressure:
not determined
Density: approx. 0,94 g/cm3 (approx. 20 °C)
Relative density: approx. 0,94 (20 °C)
Relative vapour density (air):
not applicable
Solubility in water: insoluble
Solubility (qualitative) solvent(s): organic solvents
soluble
Partitioning coefficient n-octanol/water (log Kow):
not applicable for mixtures
Self ignition: Risk of self-ignition when a large surface area is produced due to fine dispersion.

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.
Viscosity, dynamic:
not determined
Explosion hazard: Based on the chemical structure there is no indicating of explosive properties.
Fire promoting properties: not fire-propagating
Grain size distribution: The substance / product is marketed or used in a non solid or granular form.

Other information

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
When finely distributed, self-ignition is possible.

Conditions to avoid
avoid atmospheric oxygen Avoid light. Avoid heat. See MSDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Irritation

Assessment of irritating effects:
Skin contact causes slight irritation. Not irritating to the eyes.

Information on: Retinyl propionate
Experimental/calculated data:
Skin corrosion/irritation rabbit: Slightly irritating. (OECD Guideline 404)

Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available Data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:
Based on available Data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available Data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:
The substance caused malformations/developmental toxicity in laboratory animals.

Information on: Retinyl propionate
Assessment of teratogenicity:
May cause harm to the unborn child. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available Data, the classification criteria are not met.

Information on: colecalciferol; Vitamin D3
Assessment of repeated dose toxicity:
Repeated oral exposure to small quantities may affect certain organs. The substance may cause damage to the kidney even after repeated ingestion of low doses, as shown in animal studies. The substance may cause damage to blood vessels even after repeated ingestion of low doses, as shown in animal studies.

Other relevant toxicity information

The product has not been tested. The statement has been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. May cause long-term adverse effects in the aquatic environment. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Sunflower oil
Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: Retinyl propionate
Assessment of aquatic toxicity:
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. There is a high probability that the product is not acutely harmful to aquatic organisms.

Information on: colecalciferol; Vitamin D3
Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. No toxic effects occur within the range of solubility. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: Sunflower oil
Toxicity to fish:
LC50 (96 h) > 1.000 mg/l, Brachydanio rerio
The details of the toxic effect relate to the nominal concentration.

Information on: Retinyl propionate
Toxicity to fish:
LC50 (96 h) > 10.000 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
The details of the toxic effect relate to the nominal concentration.

Information on: colecalciferol; Vitamin D3
Toxicity to fish:
LC50 (96 h) > 10.000 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
The product has low solubility in the test medium. An aqueous dispersion has been tested. Tested above maximum solubility. The details of the toxic effect relate to the nominal concentration.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Retinyl propionate
Elimination information:
40 - 50 % CO2 formation relative to the theoretical value (28 d) (OECD 301B; ISO 9439; 92/69/EEC, C.4-C) (aerobic, activated sludge, domestic)

Information on: colecalciferol; Vitamin D3
Elimination information:
< 10 % BOD of the ThOD (28 d) (OECD Guideline 301 F) (aerobic, activated sludge, domestic)

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested.

Bioaccumulation potential:
Because of the product's consistency and low water solubility, bioavailability is improbable. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Retinyl propionate
Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Information on: colecalciferol; Vitamin D3
Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Information on: Sunflower oil
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is possible.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Adsorption to solid soil phase is expected. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Retinyl propionate
Assessment transport between environmental compartments:
Adsorption in soil: Adsorption to solid soil phase is expected.

Information on: colecalciferol; Vitamin D3
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is expected.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria. Self classification

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

UN number
See corresponding entries for “UN number” for the respective regulations in the tables above.

UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

Environmental hazards

See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Skin Corr./Irrit.	Skin corrosion/irritation
Repr.	Reproductive toxicity
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Acute Tox.	Acute toxicity
STOT RE	Specific target organ toxicity — repeated exposure
H316	Causes mild skin irritation.
H303	May be harmful if swallowed.
H360	May damage the unborn child.
H413	May cause long lasting harmful effects to aquatic life.
H310	Fatal in contact with skin.
H330	Fatal if inhaled.
H300	Fatal if swallowed.
H372	Causes damage to organs (Kidney, Blood vessel system) through prolonged or repeated exposure.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the

corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.